# Measure-Theoretic Probability Theory

Lecture Notes for Econometrics.

Eliab Luvanda

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# 1 Introduction

## 1.1 Probability and Uncertainty

In a world of perfect information, economic analysis would reduce to a deterministic calculus: households would allocate resources with certainty, firms would optimise production against known prices and costs, and governments would design policy in full knowledge of its consequences. The real world, however, is irreducibly uncertain. Outcomes that matter enormously to economic agents — harvests, prices, wages, health shocks, interest rates, policy changes — are not known in advance and cannot be controlled. Probability theory provides the only rigorous language for reasoning about such uncertainty. By assigning numerical measures to the likelihood of events before they are realised, probability transforms vague intuitions about chance into a precise, internally consistent calculus that can be embedded in optimisation problems, tested against data, and used to derive welfare implications. Without this foundation, the core constructs of modern economics — expected utility, risk aversion, insurance, option pricing, hypothesis testing, and identification in causal inference — would have no formal basis.

The importance of probability in economics goes beyond mere bookkeeping of likelihoods. It provides the structural scaffolding for rational decision-making under uncertainty. When an agent faces a decision whose consequences depend on an unknown state of the world, the probability space $(\Omega, \mathcal{F}, P)$ formalises the environment: $\Omega$ lists the states that might obtain, $\mathcal{F}$ specifies which events are observable and contractible, and $P$ encodes beliefs about how likely each state is. The agent's objective — typically the maximisation of expected utility — is then a probability-weighted average of utilities across states, and the entire machinery of constrained optimisation can be brought to bear. Crucially, probability also disciplines empirical work: econometric estimators, confidence intervals, hypothesis tests, and identification strategies are all defined in terms of probability limits, distributions, and measures. The probability space is therefore not an abstract luxury but the invisible infrastructure on which both economic theory and empirical practice rest.

To make these ideas concrete, consider agents operating in environments representative of the Tanzanian and broader developing-country context. The following examples illustrate how probability structures the decision problems of a smallholder farmer, a rural household, and a firm, each facing a distinct but equally fundamental form of uncertainty.

**Example 1.1** (Tanzanian Smallholder Farmer and Weather Uncertainty)**.** Consider a smallholder farmer in a village in the Dodoma region of Tanzania, preparing planting decisions ahead of the long rains season. The farmer cannot observe next season's weather in advance, but recognises three possible states of nature based on historical experience and knowledge of local climate patterns:

$$\Omega = \{\text{good weather},\ \text{flood},\ \text{drought}\},$$

with probabilities assigned on the basis of historical climatological frequencies:

$$P(\{\text{good weather}\}) = 0.55, \quad P(\{\text{flood}\}) = 0.20, \quad P(\{\text{drought}\}) = 0.25.$$

Each state maps to a distinct crop yield and hence a distinct level of household food security and income. Under good weather the farmer harvests $y_g$ kilograms of maize; under flood, $y_f < y_g$; under drought, $y_d < y_f$. The farmer's decision problem — how much land to cultivate, whether to purchase drought-resistant seed varieties, and whether to participate in a weather-indexed insurance scheme — is formally an optimisation over expected utility:

$$\max_{c,\,q} \;\; 0.55\, u(y_g - c) + 0.20\, u(y_f - c) + 0.25\, u(y_d + q - c - \pi q),$$

where $c$ denotes input expenditure, $q$ is the quantity of insurance purchased, and $\pi$ is the insurance premium per unit. Without the probability space $(\Omega, \mathcal{F}, P)$, this problem cannot even be stated, let alone solved or estimated from household survey data.

**Example 1.2** (Rural Household and Income Uncertainty)**.** Consider a rural household in Tanzania whose total income is drawn from two sources: on-farm crop income, which is subject to weather risk, and off-farm casual labour income, which fluctuates with local demand conditions. Let the household's income in state $\omega \in \Omega$ be denoted $Y(\omega)$, a random variable defined on the probability space $(\Omega, \mathcal{F}, P)$. The household's central concern is the distribution of $Y$ across states, and in particular its lower tail: the probability of falling below a subsistence threshold $\underline{y}$ is

$$P(Y < \underline{y}) = \sum_{\omega:\, Y(\omega) < \underline{y}} P(\{\omega\}),$$

which is the formal expression of poverty risk. The household manages this risk through a portfolio of strategies — precautionary saving, diversification of income sources, informal risk-sharing arrangements with neighbours, and participation in formal insurance or social protection programmes. Each of these strategies can be evaluated and ranked only by comparing the probability distributions they induce over income outcomes, making the probability space the indispensable foundation of household welfare analysis and the design of social protection policy.

**Example 1.3** (Firm and Price Uncertainty). Consider a firm operating in an agricultural output market in Tanzania, deciding how much to invest in processing capacity before the harvest price $\tilde{p}$ is realised. The price is a random variable drawn from a distribution $P$ over a Borel-measurable subset of $\mathbb{R}_+$, reflecting uncertainty about domestic supply, export demand, and government price interventions. The firm's profit in state $\omega$ is

$$\pi(\omega) = \tilde{p}(\omega) \cdot q - C(q),$$

where $q$ is the chosen capacity and $C(q)$ is the cost of that capacity. The firm maximises expected profit

$$\max_{q \geq 0} \ \mathbb{E}_P[\tilde{p}] \cdot q - C(q) = \max_{q \geq 0} \ \int_\Omega \tilde{p}(\omega)\, dP(\omega) \cdot q - C(q),$$

where the expectation is a Lebesgue integral with respect to the probability measure $P$. A risk-averse firm, by contrast, maximises the expected utility of profit, placing additional weight on unfavourable price realisations. In either case, the probability measure $P$ over price states is the primitive object of the firm's decision problem, and its accurate estimation from market data is the central task of an empirical industrial organisation or agricultural economics study. The probability space $(\Omega, \mathcal{F}, P)$ is thus not an abstract formalism but the direct structural input into firm-level investment, production, and contracting decisions.

In economics, uncertainty is central to decision-making, from consumption choices to financial markets and policy design. Probability theory provides the only rigorous language for reasoning about suchuncertainty, transforming vague intuitions about chance into a precise, internally consistent calculus that can be embedded in optimisation problems, tested against data, and used to derive welfare implications. Without this foundation, the core constructs of modern economics — expected utility, risk aversion, insurance, option pricing, hypothesis testing, and identification in causal inference — would have no formal basis.

## 1.2 Probability Theory as the Foundation of Econometric Theory

Econometrics is, at its foundational level, an application of probability theory to the empirical analysis of economic relationships. This insight was stated with remarkable clarity and force by Trygve Haavelmo in his landmark 1944 monograph *The Probability Approach in Econometrics*, for which he was awarded the Nobel Memorial Prize in Economic Sciences in 1989. Haavelmo's central argument was that economic models should not be treated as deterministic algebraic relationships fitted to data by minimising discrepancies, but should instead be conceived from the outset as *probability models* — complete specifications of the joint probability distribution of all variables entering the analysis.

On this view, the observed data are not a fixed object to be described but a single realisation drawn from a well-defined probability space $(\Omega, \mathcal{F}, P)$, and the task of econometrics is to use that realisation to learn about the probability measure $P$ that generated it. Without an explicit probabilistic framework, the concepts of estimation, consistency, efficiency, and statistical inference are simply undefined, and the applied researcher has no principled basis for distinguishing signal from noise or for quantifying the uncertainty that surrounds any empirical conclusion.

To see why the probability framework is not merely convenient but logically indispensable, consider the classical consumption function

$$C_t = \alpha + \beta Y_t + u_t,$$

where $C_t$ is household consumption in period $t$, $Y_t$ is disposable income, $\alpha$ and $\beta$ are structural parameters, and $u_t$ is an unobservable disturbance. In a purely algebraic reading, this equation is simply a line with an error attached, and estimation reduces to a curve-fitting exercise. Haavelmo's probability approach transforms this interpretation entirely. Because we work with observational rather than experimental data, income $Y_t$ is not set by the researcher but is itself generated by an underlying economic process — it is a random variable with its own marginal distribution, reflecting the aggregate interactions of labour markets, policy, technology, and household decisions that lie outside the model. Conditional on the realised value of $Y_t$ and the true but unobservable structural parameter $\beta$ — which encodes the rational marginal propensity to consume of a utility-maximising household — we observe a further random variable $C_t$, whose stochastic variation is governed by $u_t$. The full probability model is therefore a specification of the joint distribution

$$(C_t, Y_t) \sim F_{\alpha,\beta,\sigma^2},$$

parameterised by the structural parameters $(\alpha, \beta, \sigma^2)$. The population is this distribution; the sample $\{(C_t, Y_t)\}_{t=1}^{T}$ is a sequence of realisations drawn from it. Probability theory operates on the population side, characterising the data-generating process; econometrics operates on the sample side, using the observed realisations to form estimators — such as the ordinary least squares estimator $\hat{\beta}$ — and to evaluate their properties. The consistency of $\hat{\beta}$, meaning that $\hat{\beta} \xrightarrow{p} \beta$ as $T \to \infty$, is a theorem about probability limits, proved using the Law of Large Numbers. The asymptotic normality of $\hat{\beta}$, which underpins $t$-tests and confidence intervals, is a theorem about convergence in distribution, proved using the Central Limit Theorem. Every inferential statement an econometrician makes is therefore a statement about the probability model, and its validity depends entirely on whether the assumed probability space is an adequate representation of the true data-generating process.

This intimate relationship between probability theory and econometrics has a direct and consequential implication for how econometrics should be learned and practised. Econometric methods are not a self-contained toolkit that can be mastered independently of the probabilistic foundations on which they rest; they are theorems and procedures derived from, and interpretable only within, a probability-theoretic framework. A researcher who does not understand the probability space underlying a model cannot assess whether its assumptions are satisfied in a given application, cannot diagnose the source of an inconsistency when an estimator fails, and cannot evaluate whether a proposed extension or modification of a standard method is internally coherent. Conversely, a researcher with a thorough command of probability theory can reconstruct the logic of any standard estimator from first principles, detect violations of identifying assumptions by reasoning about the underlying distributions, and engage critically with the frontier of econometric theory as it evolves.

*Remark* (Probability Theory as a Necessary and Sufficient Foundation). A thorough understanding of probability theory is both a *necessary* and a *sufficient* condition for a deep understanding of econometric theory. It is necessary because every core concept in econometrics — the sampling distribution of an estimator, the consistency and asymptotic normality of extremum estimators, the size and power of hypothesis tests, the construction of confidence sets, and the identification of structural parameters from observational data — is defined and proved within a probability-theoretic framework. An econometrician who lacks this foundation is reduced to memorising formulas without understanding their scope, their assumptions, or the conditions under which they break down. It is sufficient because the entire edifice of classical and modern econometric theory is, at bottom, an application of measure-theoretic probability to the specific problem of learning from economic data: a researcher who has fully internalised the probability space, random variables, convergence concepts, and the limit theorems can derive the properties of any estimator, construct valid tests, and assess identification from first principles, without relying on the authority of textbook results.

*Remark* (Probability Theory as a Foundation for Responsible Empirical Practice). Beyond theoretical understanding, a solid grounding in probability theory is essential for the informed and responsible application of econometrics in empirical research. Many of the most consequential errors in applied economic research — inappropriate use of asymptotic approximations in small samples, failure to account for clustering or heteroskedasticity, misinterpretation of $p$-values and confidence intervals, and the conflation of statistical significance with economic significance — are not errors of computation but errors of probabilistic reasoning. A researcher who understands that a confidence interval is a statement about the sampling distribution of an estimator across hypothetical repetitions of the data-generating process, not a statement about the probability that the true parameter lies in a given range, will not misreport their results. A researcher who understands that a $p$-value is the probability, under the null hypothesis and the assumed probability model, of observing a test statistic at least as extreme as the one obtained, will not treat it as a measure of the probability that the null hypothesis is true. These distinctions are probabilistic in nature, and their mastery is a precondition for credible empirical work.

*Remark* (Haavelmo's Legacy and the Modern Structural Approach). Haavelmo's probability approach is not a historical curiosity but the living foundation of modern structural econometrics. The potential outcomes framework of Rubin and the structural approach of the Cowles Commission are both, at their core, specifications of probability models for the data-generating process under alternative interventions or counterfactuals. The identification problem — the central preoccupation of modern applied econometrics — is fundamentally a question about whether the probability distribution of observables uniquely determines the structural parameters of interest. Instrumental variables, regression discontinuity, difference-in-differences, and synthetic control methods are all strategies for using observable variation in the joint distribution of $(Y, X, Z)$ to recover parameters that are not directly identified from the marginal distribution of $Y$ alone. Understanding why any of these strategies works, and when it fails, requires exactly the probabilistic reasoning that Haavelmo placed at the centre of econometric methodology eight decades ago.

## 1.3 Approaches to Probability

### 1.3.1 Classical

The classical (or Laplacean) approach to probability defines probability in terms of equally likely outcomes within a finite sample space. Formally, let $\Omega$ be a finite sample space with

$|\Omega| < \infty$, and suppose that all elementary outcomes $\omega \in \Omega$ are equally likely. Then, for any event $A \subseteq \Omega$, the probability of $A$ is defined as

$$P(A) = \frac{|A|}{|\Omega|},$$

where $|A|$ denotes the number of favorable outcomes in $A$, and $|\Omega|$ is the total number of possible outcomes. This approach relies critically on the principle of insufficient reason (or indifference), which assigns equal probabilities to all outcomes in the absence of further information. While the classical definition provides an intuitive and mathematically convenient foundation for probability theory—particularly in games of chance such as coin tosses, dice rolls, and card draws—it is limited to settings with finite and symmetric outcome spaces, and does not readily extend to situations involving infinite sample spaces or outcomes that are not equally likely.

### 1.3.2 Frequentist

The frequentist approach to probability defines the probability of an event in terms of its long-run relative frequency in repeated, identical trials of a random experiment. Formally, consider a sequence of independent and identically distributed trials, and let $A$ be an event of interest. Let $N_n(A)$ denote the number of occurrences of $A$ in the first $n$ trials. Then, the probability of $A$ is defined as the limiting relative frequency:

$$P(A) = \lim_{n \to \infty} \frac{N_n(A)}{n},$$

provided that this limit exists. This definition is grounded in the law of large numbers, which ensures that, under suitable regularity conditions, the sample frequency converges to a stable value as the number of trials becomes large. The frequentist interpretation is particularly well-suited for empirical and experimental contexts where repeatability is meaningful; however, it is conceptually limited when dealing with unique or non-repeatable events, and it presumes the existence of an underlying stochastic mechanism generating the observed outcomes.

### 1.3.3 Subjective

The subjective (or Bayesian) approach to probability defines probability as a degree of belief or confidence that an individual assigns to the occurrence of an event, based on their personal judgment, information, and prior experience. Formally, let $A$ be an event, and let $\mathbb{P}(A)$ represent the subjective probability assigned to $A$. This interpretation is inherently personal and can vary between individuals, as it reflects their unique perspectives and information sets. The subjective approach is particularly useful in decision-making under uncertainty, where probabilities are used to guide choices in the absence of objective frequencies or symmetries. However, it can be criticized for its lack of objectivity and potential for inconsistency across different individuals.

### 1.3.4 Axiomatic (Measure-Theoretic)

Probability is defined as a measure on a measurable space.

The axiomatic (measure-theoretic) approach to probability, formalized by *Andrey Kolmogorov*, defines probability as a measure on a structured set, thereby providing a rigorous and general foundation for both finite and infinite probabilistic models. Let $(\Omega, \mathcal{F}, P)$ be a probability space, where $\Omega$ is the sample space, $\mathcal{F}$ is a $\sigma$-algebra of subsets of $\Omega$ (the collection of events), and $P : \mathcal{F} \to [0, 1]$ is a probability measure. The measure $P$ satisfies three axioms: (i) non-negativity, $P(A) \geq 0$ for all $A \in \mathcal{F}$; (ii) normalization, $P(\Omega) = 1$; and (iii) countable additivity, such that for any countable sequence of disjoint events $\{A_i\}_{i=1}^{\infty} \subset \mathcal{F}$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

This framework unifies probability with measure theory and allows probabilistic reasoning to be conducted with full mathematical rigor.

A key strength of the axiomatic approach is its ability to handle infinite and uncountable sample spaces, which arise naturally in economics and stochastic processes. For example, when $\Omega = \mathbb{R}$ (e.g., modeling continuous variables such as income, time, or prices), it is not meaningful to assign equal probability to individual outcomes. Instead, the $\sigma$-algebra $\mathcal{F}$ (typically the Borel $\sigma$-algebra) restricts attention to well-behaved subsets of $\mathbb{R}$, and probabilities are assigned via measures such as the Lebesgue measure. In this setting, probabilities are defined over intervals and more complex measurable sets, rather than points, thereby resolving the conceptual limitations of classical probability. The use of countable additivity ensures that probabilities remain consistent even when dealing with infinite collections of events, while avoiding paradoxes associated with uncountable unions that are not measurable.

The structure of a probability space can be visualized as a rectangle representing the sample space $\Omega$, whose total area is normalized to 1. The $\sigma$-algebra $\mathcal{F}$ is not a region within $\Omega$ but rather a *collection of admissible subsets* of $\Omega$ — those subsets to which the probability measure $P$ is permitted to assign a value. An event $A \in \mathcal{F}$ is one such admissible subset, depicted as the shaded region, and its probability $P(A)$ corresponds to the ratio of the shaded area to the total area of $\Omega$:

This diagram highlights that $P(A)$ is always measured relative to the whole of $\Omega$, consistently with the requirement $P(\Omega) = 1$. The role of $\mathcal{F}$ is not to carve out a sub-region of $\Omega$ but to specify which subsets are *measurable* — a distinction that becomes essential when $\Omega$ is uncountable, since not every subset of an uncountable space can be assigned a probability in a logically consistent way. By embedding probability within measure theory, the axiomatic approach provides a coherent and flexible framework capable of supporting modern econometrics, stochastic calculus, and advanced statistical inference.

## 2 Probability Spaces

A probability space is the fundamental mathematical structure that underlies all of modern probability theory, providing a rigorous foundation for reasoning about uncertainty. Formally

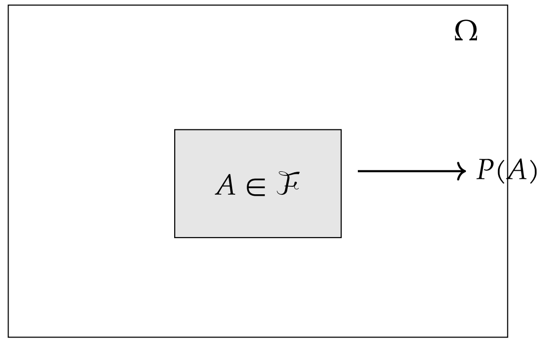


Figure 1: Structure of a probability space $(\Omega, \mathcal{F}, P)$. The outer rectangle represents the sample space $\Omega$, normalized so that its total area equals 1. The shaded region is an event $A \in \mathcal{F}$, and $P(A)$ is the ratio of the shaded area to the total area of $\Omega$. The $\sigma$-algebra $\mathcal{F}$ is not depicted as a region but is understood as the family of all subsets of $\Omega$ — such as $A$ — to which $P$ can be legitimately assigned.

introduced by Andrei Kolmogorov in his 1933 *Grundbegriffe der Wahrscheinlichkeitsrechnung*, it is defined as a triple $(\Omega, \mathcal{F}, P)$ consisting of three objects that work in concert: a sample space $\Omega$, a $\sigma$-algebra $\mathcal{F}$, and a probability measure $P$. The power of this construction lies not in any single component but in the disciplined way the three interact — $\Omega$ specifies what can happen, $\mathcal{F}$ specifies what can be meaningfully measured, and $P$ assigns to each measurable event a number in $[0, 1]$ that quantifies its likelihood. Together, they elevate probability from an informal intuition about chance to a branch of measure theory, granting it the full deductive machinery of mathematical analysis and enabling a unified treatment of both discrete and continuous phenomena within a single, coherent framework.

## 2.1 Sample Space

The sample space is the starting point of any probabilistic model: before assigning likelihoods to outcomes, one must first declare precisely what outcomes are possible. Intuitively, it is an exhaustive list of everything that could happen in a given experiment or situation, with no ambiguity about what constitutes a distinct outcome.

**Definition 2.1** (Sample Space)**.** The **sample space**, denoted $\Omega$, is the set of all possible outcomes of a random experiment. Each element $\omega \in \Omega$ is called a **sample point** or **elementary outcome**, and the collection must be mutually exclusive and collectively exhaustive — that is, exactly one $\omega \in \Omega$ will be realized on any given run of the experiment.

The two conditions embedded in this definition carry real logical weight. Mutual exclusivity requires that the outcomes be defined finely enough that no two can occur simultaneously: if two outcomes overlap, the model is underspecified. Collective exhaustiveness requires that the sample space be defined broadly enough to contain every conceivable realization: if an outcome occurs that is not in $\Omega$, the model is incomplete. In practice, the analyst constructs $\Omega$ by a deliberate act of abstraction — retaining only those features of reality that are relevant to the question at hand and discarding the rest. A sample space may be finite, countably infinite, or uncountable, and this distinction has profound consequences for the subsequent mathematical machinery required to assign probabilities.

**Example 2.2** (Tossing a Coin)**.** A fair coin is tossed once. The two possible outcomes are a head and a tail, so the sample space is

$$\Omega = \{\mathrm{H}, \mathrm{T}\}.$$

This is the simplest non-trivial sample space: finite, with two mutually exclusive and collectively exhaustive elementary outcomes.

**Example 2.3** (Tossing a Die)**.** A standard six-sided die is rolled once. Each face shows a distinct number of pips, yielding six possible outcomes, so the sample space is

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Any event of practical interest — such as rolling an even number — corresponds to a subset of $\Omega$, for instance $A = \{2, 4, 6\}$.

**Example 2.4** (Weather Conditions and Farm Output)**.** A farmer operates in a region subject to three mutually exclusive and collectively exhaustive seasonal weather regimes: favorable conditions, flooding, and drought. From the perspective of an econometrician modeling agricultural output or crop insurance take-up, the relevant sample space is

$$\Omega = \{\text{good weather}, \ \text{flood}, \ \text{drought}\}.$$

Each $\omega \in \Omega$ triggers a distinct production outcome — and hence a distinct realization of farm revenue — making $\Omega$ the primitive object over which a structural model of decision-making under uncertainty is built. Extensions of this framework allow each weather state to be associated with a continuous yield distribution, in which case $\Omega$ would be enriched to accommodate a continuum of possible output realizations.

## 2.2 Sigma-Algebra

Where the sample space $\Omega$ catalogues every possible outcome of a random experiment, the $\sigma$-algebra refines the model by specifying precisely which *collections* of outcomes are admissible objects of probabilistic reasoning. Not every subset of $\Omega$ need be — or, in uncountable spaces, can be — assigned a probability in a logically consistent way, and the $\sigma$-algebra is the mathematical device that enforces this discipline.

**Definition 2.5** ($\sigma$-Algebra)**.** Let $\Omega$ be a sample space. A **$\sigma$-algebra** (or $\sigma$-field) on $\Omega$ is a collection $\mathcal{F}$ of subsets of $\Omega$ satisfying the following three axioms:

1. **Contains the sample space:** $\Omega \in \mathcal{F}$.
2. **Closed under complementation:** If $A \in \mathcal{F}$, then $A^c = \Omega \setminus A \in \mathcal{F}$.
3. **Closed under countable unions:** If $A_1, A_2, A_3, \ldots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

The pair $(\Omega, \mathcal{F})$ is called a **measurable space**, and any set $A \in \mathcal{F}$ is called a **measurable event**.

Each axiom has a precise logical purpose. The first ensures that the certain event — something happens — is always assignable a probability, namely $P(\Omega) = 1$. The second ensures coherence under negation: if one can speak meaningfully about the probability of an event $A$ occurring, one must equally be able to speak about the probability of $A$ *not* occurring, and that complement must also be a measurable event. The third axiom extends this coherence to countably infinite sequences of events, ensuring that asking whether *at least one* of a sequence of events occurs is always a well-posed question. Taken together, the three axioms guarantee that $\mathcal{F}$ is closed under all the set-theoretic operations — complementation, countable union, and hence countable intersection by De Morgan's laws — that arise naturally in probabilistic arguments.

*Remark* (Why the Sample Space Alone Is Not Enough). One might reasonably ask why $\Omega$ is insufficient on its own and why a $\sigma$-algebra is needed at all. Consider the simple experiment of tossing a fair coin, where $\Omega = \{\mathrm{H}, \mathrm{T}\}$. Here $\Omega$ is finite, and there is no difficulty in assigning probabilities to every subset of $\Omega$: the power set $2^{\Omega} = \{\emptyset, \{\mathrm{H}\}, \{\mathrm{T}\}, \Omega\}$ serves perfectly well as the $\sigma$-algebra, and one can assign $P(\{\mathrm{H}\}) = P(\{\mathrm{T}\}) = \frac{1}{2}$ without any contradiction. In this finite setting, the $\sigma$-algebra is a formality. The necessity becomes acute when $\Omega$ is uncountable — as it is whenever outcomes are modeled as real numbers, for instance when a crop yield, an asset return, or a wage is drawn from a continuous distribution. In such cases, it can be shown (via constructions such as the Vitali set) that no probability measure can be defined consistently on *all* subsets of $\Omega$; some subsets are simply not measurable. The $\sigma$-algebra solves this problem by restricting attention to a well-behaved subfamily of subsets — large enough to contain every event of practical interest, but small enough to admit a coherent probability measure. For the real line, this subfamily is the Borel $\sigma$-algebra $\mathcal{B}(\mathbb{R})$, generated by all open intervals, which contains all intervals, open and closed sets, and every set an econometrician is ever likely to encounter.

**Example 2.6** ($\sigma$-Algebra for a Coin Toss)**.** Let $\Omega = \{\mathrm{H}, \mathrm{T}\}$. The unique $\sigma$-algebra that contains all subsets of $\Omega$ is the power set

$$\mathcal{F} = \{\emptyset,\ \{\mathrm{H}\},\ \{\mathrm{T}\},\ \Omega\}.$$

One can verify all three axioms directly: $\Omega \in \mathcal{F}$; the complement of each set is also in $\mathcal{F}$; and every union of elements of $\mathcal{F}$ remains in $\mathcal{F}$. In this finite setting the power set is the natural and only sensible choice of $\sigma$-algebra, and it permits probabilities to be assigned to every conceivable event.

**Example 2.7** ($\sigma$-Algebra for a Die Roll)**.** Let $\Omega = \{1, 2, 3, 4, 5, 6\}$. The power set $2^{\Omega}$ contains $2^6 = 64$ subsets and constitutes the maximal $\sigma$-algebra on $\Omega$, allowing probabilities to be assigned to every event. A strictly coarser $\sigma$-algebra can be constructed by grouping outcomes: for instance, defining $E = \{2, 4, 6\}$ (even) and $O = \{1, 3, 5\}$ (odd) yields

$$\mathcal{F} = \{\emptyset,\ E,\ O,\ \Omega\},$$

which is a valid $\sigma$-algebra that supports only the question of whether the outcome is even or odd, discarding all finer information about the individual face. This illustrates that the choice of $\sigma$-algebra encodes the *information structure* of the model.

**Example 2.8** ($\sigma$-Algebra for Weather States)**.** Let the sample space be

$$\Omega = \{\text{good}, \ \text{flood}, \ \text{drought}\},$$

representing mutually exclusive and exhaustive seasonal weather outcomes faced by a farmer. Define the $\sigma$-algebra $\mathcal{F}$ as the power set of $\Omega$, i.e.

$$\mathcal{F} = 2^{\Omega} = \big\{\emptyset, \quad \{\text{good}\}, \{\text{flood}\}, \quad \{\text{drought}\}, \{\text{good}, \text{flood}\}, \quad \{\text{good}, \text{drought}\}, \{\text{flood}, \text{drought}\}, \quad \Omega\big\}.$$

Then $(\Omega, \mathcal{F})$ is a measurable space. Any probability measure $P : \mathcal{F} \to [0, 1]$ satisfying $P(\Omega) = 1$ completes the probability space $(\Omega, \mathcal{F}, P)$.

*Interpretation.* Each element $A \in \mathcal{F}$ represents a well-defined event. For example,

$$P(\{\text{flood}, \text{drought}\})$$

is the probability of an adverse weather outcome. In economic terms, such events may define trigger conditions for crop insurance contracts or government relief programs.

ample, [ P({flood, drought}) ] is the probability of an adverse weather outcome. In economic terms, such events may define trigger conditions for crop insurance contracts or government relief programs. \end{example}

*Remark.* In a finite sample space, the $\sigma$-algebra generated by $\Omega$ is typically taken to be the full power set, ensuring that *all* subsets are measurable. This guarantees that any economically meaningful event — whether simple (e.g. "flood") or composite (e.g. "not good weather") — can be assigned a probability. In more general (infinite) settings, however, not every subset of $\Omega$ can be assigned a probability in a logically consistent way: such subsets are said to be extit{non-measurable}, and no $\sigma$-algebra can contain them without destroying the coherence of the probability measure. The $\sigma$-algebra therefore delineates precisely the boundary between sets that are amenable to probabilistic measurement and those that are not — the latter being not merely excluded by convention but genuinely uncontrollable within any coherent probability framework.

## 2.3 Probability Measure

With the sample space $\Omega$ enumerating all possible outcomes and the $\sigma$-algebra $\mathcal{F}$ identifying which events are admissible objects of measurement, the final ingredient of a probability space is the probability measure $P$ — the function that completes the triple $(\Omega, \mathcal{F}, P)$ by assigning to each measurable event a number that quantifies its likelihood. The probability measure is not defined arbitrarily; it must satisfy a set of axioms, known as the Kolmogorov axioms, that encode the minimal logical requirements for a coherent calculus of chance.

**Definition 2.9** (Probability Measure)**.** Let $(\Omega, \mathcal{F})$ be a measurable space. A **probability measure** is a function $P : \mathcal{F} \to [0, 1]$ satisfying the following three axioms:

1. **Non-negativity:** For every event $A \in \mathcal{F}$,

$$P(A) \geq 0.$$

2. **Normalization:** The probability of the certain event is one,

$$P(\Omega) = 1.$$

3. **Countable additivity ($\sigma$-additivity):** For any countable collection of mutually exclusive events $A_1, A_2, A_3, \ldots \in \mathcal{F}$ satisfying $A_i \cap A_j = \emptyset$ for all $i \neq j$,

$$P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i).$$

The triple $(\Omega, \mathcal{F}, P)$ is called a **probability space**.

Each axiom distills a fundamental requirement of rational belief about uncertainty. Non-negativity reflects the irreducibly non-negative nature of likelihood: no event can be less probable than impossible, and impossibility itself is represented by $P(\emptyset) = 0$, which follows immediately from the other axioms. Normalization anchors the scale: by convention, certainty is assigned the value one, so that all probabilities are expressed as fractions of the whole. It is this axiom that gives the geometric interpretation its force — $P(A)$ is literally the proportion of the total probability mass, normalized to unity, that is allocated to $A$. Countable additivity is the most substantive of the three: it asserts that when events are mutually exclusive, their probabilities *add*, and that this additivity extends to countably infinite collections of events, not merely finite ones. This extension to the countably infinite case is what distinguishes $\sigma$-additivity from the weaker requirement of finite additivity, and it is precisely what enables the passage from sums to integrals and from discrete to continuous probability distributions. Together the three axioms imply all the familiar rules of probability — the complement rule $P(A^c) = 1 - P(A)$, the addition rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, and monotonicity $A \subseteq B \Rightarrow P(A) \leq P(B)$ — as purely logical consequences, requiring no further assumptions.

**Example 2.10** (Probability Measure for a Coin Toss)**.** Let $\Omega = \{\mathrm{H}, \mathrm{T}\}$ and $\mathcal{F} = \{\emptyset, \{\mathrm{H}\}, \{\mathrm{T}\}, \Omega\}$. For a fair coin, the probability measure $P$ is defined by assigning

$$P(\emptyset) = 0, \quad P(\{\mathrm{H}\}) = \tfrac{1}{2}, \quad P(\{\mathrm{T}\}) = \tfrac{1}{2}, \quad P(\Omega) = 1.$$

One verifies immediately that all three Kolmogorov axioms are satisfied: probabilities are non-negative, they sum to one over the whole space, and because $\{\mathrm{H}\}$ and $\{\mathrm{T}\}$ are mutually exclusive, $P(\{\mathrm{H}\} \cup \{\mathrm{T}\}) = P(\{\mathrm{H}\}) + P(\{\mathrm{T}\}) = 1 = P(\Omega)$, as required.

**Example 2.11** (Probability Measure for a Die Roll)**.** Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ and let $\mathcal{F} = 2^{\Omega}$ be the full power set. For a fair die, each elementary outcome is assigned equal probability:

$$P(\{\omega\}) = \frac{1}{6} \quad \text{for each } \omega \in \Omega.$$

By $\sigma$-additivity, the probability of any event $A \subseteq \Omega$ is

$$P(A) = \frac{|A|}{6},$$

where $|A|$ denotes the cardinality of $A$. For instance, the probability of rolling an even number is $P(\{2,4,6\}) = \frac{3}{6} = \frac{1}{2}$, and the probability of rolling a number greater than four is $P(\{5,6\}) = \frac{2}{6} = \frac{1}{3}$.

**Example 2.12** (Probability Measure for Weather States)**.** Let $\Omega = \{\text{good, flood, drought}\}$ and let $\mathcal{F} = 2^{\Omega}$ be the full power set. Suppose that historical agronomic data for a given region yield the following climatological frequencies:

$$P(\{\text{good}\}) = 0.60, \quad P(\{\text{flood}\}) = 0.25, \quad P(\{\text{drought}\}) = 0.15.$$

These values are non-negative and sum to one, satisfying the Kolmogorov axioms. By $\sigma$-additivity, the probability of any compound event is obtained by summing over its constituent states; for example, the probability of an adverse weather realization — the event that triggers a crop insurance payout — is

$$P(\{\text{flood, drought}\}) = P(\{\text{flood}\}) + P(\{\text{drought}\}) = 0.25 + 0.15 = 0.40.$$

An econometrician estimating a structural model of farmer behavior under uncertainty would treat these probabilities as primitives of the decision problem, using them to compute expected utilities, optimal insurance coverage, and the welfare effects of weather-indexed transfer programs.

The three Kolmogorov axioms are not merely definitional conveniences; they are sufficiently powerful to imply a rich set of properties that govern the behaviour of any probability measure. The following theorem collects the four most fundamental of these derived properties.

**Theorem 2.13** (Derived Properties of a Probability Measure)**.** *Let $(\Omega, \mathcal{F}, P)$ be a probability space. Then the following properties hold for all $A, B \in \mathcal{F}$:*

1. ***Probability of the empty set:***
$$P(\emptyset) = 0.$$

2. ***Complement rule:***
$$P(A^c) = 1 - P(A).$$

3. ***Monotonicity:***
$$A \subseteq B \implies P(A) \leq P(B).$$

4. ***Addition rule (inclusion-exclusion):***
$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

*Proof.* **Property 1.** Since $\Omega$ and $\emptyset$ are mutually exclusive and $\Omega \cup \emptyset = \Omega$, countable additivity gives
$$P(\Omega) = P(\Omega \cup \emptyset) = P(\Omega) + P(\emptyset).$$
Subtracting $P(\Omega)$ from both sides yields $P(\emptyset) = 0$. $\square$

**Property 2.** For any $A \in \mathcal{F}$, the sets $A$ and $A^c$ are mutually exclusive and satisfy $A \cup A^c = \Omega$. By countable additivity,
$$P(A \cup A^c) = P(A) + P(A^c).$$
Since $P(\Omega) = 1$ by normalization, it follows immediately that
$$P(A) + P(A^c) = 1 \implies P(A^c) = 1 - P(A). \quad \square$$

**Property 3.** Suppose $A \subseteq B$. Then $B$ can be decomposed into two mutually exclusive sets:
$$B = A \cup (B \setminus A), \quad A \cap (B \setminus A) = \emptyset.$$
By countable additivity,
$$P(B) = P(A) + P(B \setminus A).$$
Since $P(B \setminus A) \geq 0$ by non-negativity, it follows that $P(A) \leq P(B)$. $\square$

**Property 4.** Any union $A \cup B$ can be written as the union of three mutually exclusive sets:
$$A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A).$$
By countable additivity,
$$P(A \cup B) = P(A \setminus B) + P(A \cap B) + P(B \setminus A).$$
Observe that $A = (A \setminus B) \cup (A \cap B)$ and $B = (B \setminus A) \cup (A \cap B)$ are also disjoint decompositions, so
$$P(A) = P(A \setminus B) + P(A \cap B), \quad P(B) = P(B \setminus A) + P(A \cap B).$$
Adding these two expressions and substituting back gives
$$P(A) + P(B) = P(A \setminus B) + P(B \setminus A) + 2\,P(A \cap B) = P(A \cup B) + P(A \cap B),$$
from which
$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \quad \square$$

■

Each of these four properties illuminates a distinct facet of probabilistic reasoning. Property~1 establishes that impossibility is represented by zero probability: the empty set $\emptyset$ contains no outcomes, so no probability mass can be allocated to it, and this follows as a theorem rather than an additional assumption. Property~2 formalises the intuition that certainty is conserved: since $A$ and its complement $A^c$ together account for all possibilities, their probabilities must sum to one, and knowing $P(A)$ immediately determines $P(A^c)$. This is one of the most frequently used tools in applied probability, since it is often far easier to compute the probability of an event *not* occurring and subtract from one than to compute its probability directly.

Property~3 encodes a basic requirement of logical consistency: if every outcome in $A$ is also an outcome in $B$, then $B$ is at least as likely as $A$, and the probability measure must respect this ordering. In econometric applications, monotonicity underpins the coherence of nested hypothesis tests and the nesting of information sets in dynamic models. Property~4, the inclusion-exclusion principle, corrects for double-counting: when two events overlap, summing their individual probabilities counts the intersection twice, and subtracting $P(A \cap B)$ restores the correct total. This property generalises to arbitrarily many events and is the foundation of combinatorial probability calculations, as well as of the union bound widely used in asymptotic theory and statistical learning.

## 2.4 Continuity Properties of Probability Measures

Imagine watching a sequence of events slowly shrink or expand toward some limiting event. For instance, picture a nested sequence of intervals on the real line, each one contained in the previous, all collapsing toward a single point. Intuitively, the probability of landing in any one of those intervals should gradually approach the probability of landing exactly at the limiting point. This natural and reassuring behavior—that probabilities *follow* their events to the limit—is precisely what the **Continuity of Probability** theorem guarantees. Far from being an abstract technicality, this result is the bridge between the combinatorial, finite world where probability is easy to compute and the analytic, infinite world where calculus and limits live. It tells us that a probability measure, much like a continuous function, does not jump erratically: as sets converge, so do their probabilities. This idea can be formally stated in a proposition.

**Proposition 2.14** (Continuity of Probability)**.** *Let* $(\Omega, \mathcal{F}, P)$ *be a probability space.*

*extbf*$(i)$ *Continuity from below. If* $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$ *is an increasing sequence of events in* $\mathcal{F}$*, then*

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} P(A_n).$$

*extbf*$(ii)$ *Continuity from above. If* $A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$ *is a decreasing sequence of events in* $\mathcal{F}$*, then*

$$P\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} P(A_n).$$

*Proof.* extbf(i) Continuity from below. Define disjoint sets by ”peeling off” successive layers:

$$B_1 := A_1, \qquad B_n := A_n \setminus A_{n-1} \quad \text{for } n \geq 2.$$

By construction the $B_n$ are pairwise disjoint, each $B_n \in \mathcal{F}$, and

$$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n, \qquad \bigcup_{k=1}^{n} B_k = A_n \quad \text{for all } n.$$

Applying countable additivity of $P$ and then finite additivity yields

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = P\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} P(B_n) = \lim_{n\to\infty} \sum_{k=1}^{n} P(B_k) = \lim_{n\to\infty} P\left(\bigcup_{k=1}^{n} B_k\right) = \lim_{n\to\infty} P(A_n).$$

extbf(ii) Continuity from above. Consider the complementary sequence $A_n^c$. Since $A_1 \supseteq A_2 \supseteq \cdots$, we have $A_1^c \subseteq A_2^c \subseteq \cdots$, an increasing sequence. By part (i):

$$P\left(\bigcup_{n=1}^{\infty} A_n^c\right) = \lim_{n\to\infty} P(A_n^c).$$

By De Morgan's law, $\bigcup_{n=1}^{\infty} A_n^c = \left(\bigcap_{n=1}^{\infty} A_n\right)^c$. Using complementation $P(E^c) = 1 - P(E)$:

$$1 - P\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty}(1 - P(A_n)) = 1 - \lim_{n\to\infty} P(A_n),$$

and rearranging gives the result. ■

### 2.4.1 Intuition and Logic

The deep idea here is that a probability measure behaves like a *flow of mass*: every outcome $\omega \in \Omega$ carries a tiny parcel of probability, and when we accumulate outcomes into a set we collect the corresponding mass. For an increasing sequence of sets, each new set merely *adds* more outcomes and hence more mass; the running total $P(A_n)$ is monotone non-decreasing and bounded above by 1, so it must converge. The key algebraic trick in the proof—decomposing the union into disjoint "rings'' $B_n$—lets us cash in countable additivity, the cornerstone axiom, and turn an infinite union into a convergent series. Continuity from above follows almost for free by duality: complementation flips a decreasing family into an increasing one, and we recycle the first result. Together, the two statements say that the probability measure $P$ is *sequentially continuous* with respect to monotone set limits, in perfect analogy with how a continuous real-valued function commutes with limits of sequences.

*Remark* (Necessity of the finiteness condition). In continuity from above, the hypothesis $P(A_1) < \infty$ is automatically satisfied since we are in a probability space ($P(\Omega) = 1$). In the more general setting of $\sigma$-finite measures, the analogous result *requires* $P(A_1) < \infty$; without it, the statement can fail. A classical counterexample in measure theory is $A_n = [n, \infty)$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$ with Lebesgue measure: $\lambda(A_n) = \infty$ for all $n$, yet $\bigcap_n A_n = \emptyset$ has measure zero.

*Remark* (Equivalent formulation via $\sigma$-additivity). Continuity from below is in fact *equivalent* to countable additivity given finite additivity and the other Kolmogorov axioms. Some axiomatic treatments therefore adopt continuity from below as the third axiom in place of $\sigma$-additivity, yielding an equivalent but sometimes more geometrically transparent foundation.

*Remark* (Subadditivity and Borel–Cantelli). Continuity of probability underpins many classical results. For instance, the first Borel–Cantelli lemma—which states that if $\sum_n P(A_n) < \infty$ then $P(A_n \text{ i.o.}) = 0$—relies on writing the limsup event as a decreasing intersection of unions and then applying continuity from above to pass the limit through $P$.

*Remark* (Topological perspective). From a functional-analytic viewpoint, the space of probability measures on a metric space can itself be equipped with the topology of weak convergence. Continuity of probability for monotone set sequences is a special case of the more general principle that integrals (and hence probabilities) behave continuously under dominated convergence, a result that ultimately rests on the same $\sigma$-additivity arguments presented above.

# 3 Random Variables

Recall that a probability space is a triple $(\Omega, \mathcal{F}, P)$, where $\Omega$ is the *sample space* (the set of all possible outcomes), $\mathcal{F}$ is a $\sigma$-algebra of subsets of $\Omega$ (the collection of events to which we can assign probabilities), and $P : \mathcal{F} \to [0,1]$ is a probability measure satisfying Kolmogorov's axioms. In many applications, we are not directly interested in the raw outcomes $\omega \in \Omega$ but rather in some numerical quantity that depends on the outcome. A *random variable* $X$ is a rule that assigns a number to each outcome—a bridge from the abstract probability space into the real line $\mathbb{R}$, where standard analytical tools become available.

**Definition 3.1** (Random Variable). Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be the measurable space consisting of the real line equipped with its Borel $\sigma$-algebra. A **random variable** is a function

$$X : \Omega \longrightarrow \mathbb{R}$$

that is $\mathcal{F}$*-measurable*, meaning that for every Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$X^{-1}(B) \;=\; \{\, \omega \in \Omega : X(\omega) \in B \,\} \;\in\; \mathcal{F}.$$

Equivalently, $X$ is a random variable if and only if

$$\{\, \omega \in \Omega : X(\omega) \leq x \,\} \;\in\; \mathcal{F} \qquad \text{for every } x \in \mathbb{R}.$$

The definition may appear technical at first, but its logic is both natural and necessary. The measurability condition says that for any numerical range of interest $B$—for instance, the outcome produces a value no greater than~5—the set of raw outcomes $\omega$ that lead to a value in $B$ must be an *event*, that is, a member of $\mathcal{F}$. This is precisely the requirement needed for the statement $X$ lies in $B$ to have a well-defined probability: we need the pre-image $X^{-1}(B)$ to live inside $\mathcal{F}$ so that $P$ can be applied to it. Without measurability, the expression $P(X \in B)$ would be meaningless—we would be asking for the probability of a set that the probability measure cannot evaluate. In short, a random variable is not "random'' in the colloquial sense of being haphazard; it is a perfectly deterministic function. What is random is the input $\omega$, drawn according to $P$. The randomness of the outcome flows through the function $X$ to produce a random numerical value.

**Example 3.2** (Tossing a fair coin.). Let $\Omega = \{H, T\}$, $\mathcal{F} = 2^{\Omega}$, and $P(H) = P(T) = \frac{1}{2}$. Define $X : \Omega \to \mathbb{R}$ by $X(H) = 1$ and $X(T) = 0$. Then $X$ is a random variable that encodes "heads" as 1 and "tails" as 0. Every subset of $\Omega$ is in $\mathcal{F}$, so measurability is trivially satisfied. $X$ follows a Bernoulli($\frac{1}{2}$) distribution.

**Example 3.3** (Tossing a fair die)**.** Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = 2^{\Omega}$, and $P(\{\omega\}) = \frac{1}{6}$ for each $\omega$. The identity function $X(\omega) = \omega$ is a random variable mapping each face to its numerical value. Again measurability holds trivially. More interesting random variables on the same space include $Y(\omega) = 1[\omega \text{ is even}]$ (indicator of an even outcome) or $Z(\omega) = \omega^2$.

**Example 3.4** (A smallholder farmer in Tanzania)**.** Consider a small-scale farmer in a village facing uncertain rainfall during the growing season. Let $\Omega$ be the set of all possible weather scenarios over the season (combinations of rainfall amounts, timing, and temperature). Define $\mathcal{F}$ as a suitable $\sigma$-algebra of weather events and $P$ as the probability measure reflecting climatological knowledge. The farmer's harvest yield $Y$ (in kilograms) is a function of the realised weather scenario: $Y(\omega)$ is the harvest that results when weather scenario $\omega$ occurs. Since $Y$ maps uncertain environmental outcomes to real numbers and the event $\{Y \leq y\}$ is measurable for each $y$, $Y$ is a random variable. The farmer faces not just a single unknown number but an entire *distribution* of possible yields induced by $P$ through $Y$.

**Example 3.5** (A household)**.** Consider a household whose monthly consumption expenditure depends on random factors such as income shocks, health events, and price fluctuations. Let $\Omega$ be the space of all possible states of the world affecting the household, equipped with an appropriate $\sigma$-algebra $\mathcal{F}$ and probability measure $P$. The household's consumption expenditure $C(\omega)$ is a random variable mapping each state of the world to a non-negative real number. Events such as $\{C(\omega) \leq c\}$—the event that the household spends no more than $c$—must be in $\mathcal{F}$ for $C$ to qualify as a proper random variable, ensuring that we can compute $P(C \leq c)$, the probability that consumption falls at or below a given threshold.

**Example 3.6** (A firm)**.** A firm produces output whose quantity and price depend on uncertain demand conditions, input costs, and productivity shocks. Let $\omega \in \Omega$ represent a realised state of the business environment. The firm's profit

$$\Pi(\omega) \; = \; p(\omega)\, q(\omega) \; - \; c(\omega)$$

is a function of random prices $p$, quantities $q$, and costs $c$—each themselves random variables on the same probability space. The composite function $\Pi$ is also a random variable (since measurable functions are closed under algebraic operations), enabling the analysis of expected profit, profit volatility, and the probability of loss.

## 3.1 Random Variables as Measurable Functions

It is worth pausing to unpack what it means, in plain terms, to say that $X$ is a *measurable function*. Measurability is simply a *compatibility* requirement between two structures: the $\sigma$-algebra $\mathcal{F}$ on the domain $\Omega$ (which says which subsets of outcomes are observable events) and the Borel $\sigma$-algebra $\mathcal{B}(\mathbb{R})$ on the codomain $\mathbb{R}$ (which says which subsets of numbers are well-behaved sets). The condition $X^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{B}(\mathbb{R})$ says precisely: whenever you identify a "nice" numerical set $B$ on the output side, pulling it back through $X$ gives you a "nice" event on the input side. The function *respects* the observable structure.

A useful way to build intuition is to think about what *fails* when measurability is violated. If $X^{-1}(B) \notin \mathcal{F}$ for some $B$, it means there is a perfectly sensible numerical question—does $X$ land in $B$—for which no probability can be assigned. The probability measure $P$ is only defined on $\mathcal{F}$, so if the pre-image falls outside $\mathcal{F}$, the question is literally unanswerable within the model. Measurability ensures this never happens: every numerical question about $X$ translates back into an event whose probability is defined. In practice, all functions encountered in economics and finance—continuous functions, step functions, limits of sequences of measurable functions—are automatically measurable with respect to standard $\sigma$-algebras, so measurability is rarely a binding constraint in applications. Its importance is foundational: it guarantees that the induced distribution guarantees that the induced distribution $P_X(B) \equiv P(X^{-1}(B))$, known as the *pushforward measure* or *law* of $X$, is itself a well-defined probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. In this sense, a random variable does not merely "produce'' random numbers—it *transports* the entire probability structure from the abstract sample space to the real line.

### 3.2 Distribution Function

**Definition 3.7.** The distribution function is

$$F_X(x) = \mathbb{P}(X \leq x)$$

---

## 4 4. Distributions

### 4.1 4.1 CDF Properties

**Theorem 4.1.** *Every CDF is non-decreasing, right-continuous, and satisfies:*

$$\lim_{x \to -\infty} F(x) = 0, \quad \lim_{x \to \infty} F(x) = 1$$

### 4.2 4.2 Independence

**Definition 4.2.** Random variables $X, Y$ are independent if

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$$

---

## 5 5. Expectations

### 5.1 5.1 Expected Value

**Definition 5.1.** For a random variable $X$,

$$\mathbb{E}[X] = \int X d\mathbb{P}$$

## 5.2 5.2 Properties

**Theorem 5.2.** *Expectation is linear:*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

*Proof.* Follows directly from properties of the Lebesgue integral. ∎

## 5.3 5.3 Jensen's Inequality

**Theorem 5.3.** *If $\phi$ is convex, then*

$$\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]$$

---

# 6 6. Convergence Concepts

## 6.1 6.1 Almost Sure Convergence

**Definition 6.1.** $X_n \to X$ almost surely if

$$\mathbb{P}(\lim X_n = X) = 1$$

## 6.2 6.2 Convergence in Probability

**Definition 6.2.** $X_n \to X$ in probability if

$$\mathbb{P}(|X_n - X| > \varepsilon) \to 0$$

## 6.3 6.3 Convergence in Distribution

**Definition 6.3.** $X_n \to X$ in distribution if

$$F_{X_n}(x) \to F_X(x)$$

## 6.4 6.4 Relationships

**Theorem 6.4.** *Almost sure convergence implies convergence in probability.*

---

## 6.5 6.5 Weak Law of Large Numbers

**Theorem 6.5.** *Let $X_i$ be i.i.d. with finite mean $\mu$. Then*

$$\bar{X}_n \to \mu \text{ in probability}$$

## 6.6 6.6 Central Limit Theorem

**Theorem 6.6.** *Let $X_i$ be i.i.d. with mean $\mu$ and variance $\sigma^2$. Then*

$$\sqrt{n}(\bar{X}_n - \mu) \to N(0, \sigma^2)$$

---

# 7 7. Exercises

**Exercise 7.1.** Prove that a sigma-algebra is closed under finite unions.

**Exercise 7.2.** Show that independence implies zero covariance, but not vice versa.

**Exercise 7.3.** Prove Jensen's inequality for discrete random variables.

**Exercise 7.4.** Simulate the Law of Large Numbers in R.

```r
library(ggplot2)
set.seed(123)
x <- rnorm(1000)
means <- cumsum(x)/seq_along(x)
df <- data.frame(n = seq_along(means), mean = means)
ggplot(df, aes(x = n, y = mean)) +
  geom_line() +
  theme_classic() +
  labs(x = "Sample size", y = "Mean")
```

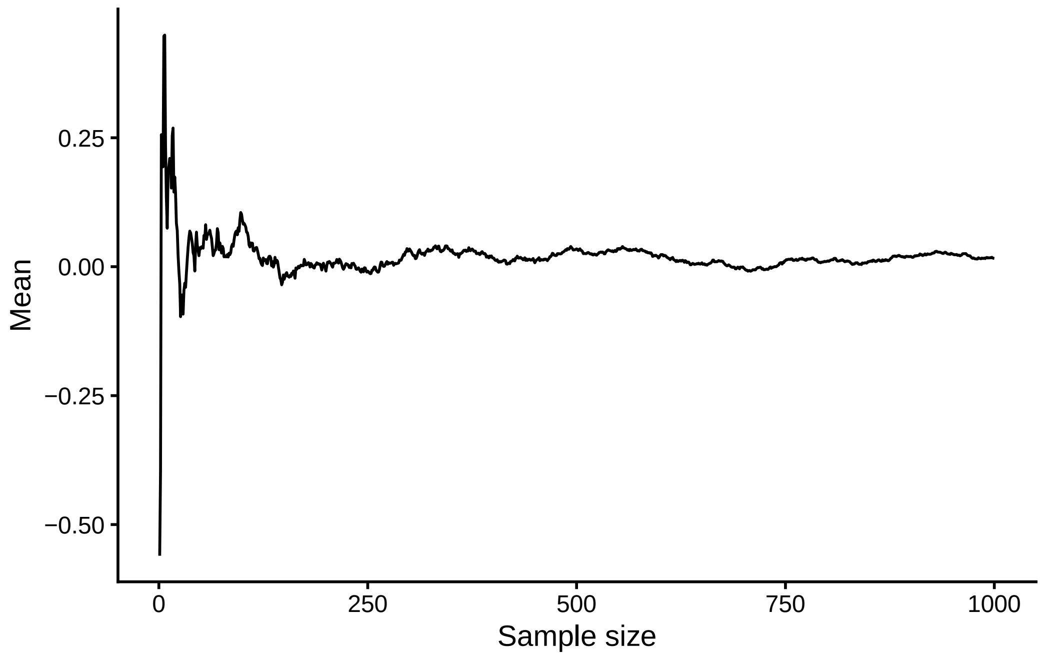


Figure 2: Simulating the Law of Large Numbers

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# 8 Appendix

## 8.1 Lebesgue Measure and Lebesgue Integration

This appendix is a self-contained, intuition-first introduction to two ideas that underlie modern probability theory: **Lebesgue measure** and the **Lebesgue integral**. No prior exposure to real analysis is assumed. The goal is not a complete treatment— proofs of the harder theorems are omitted—but rather to give you enough geometric and conceptual grounding to read measure-theoretic probability with confidence.

### 8.1.1 Why Do We Need a New Notion of "Length"?

You already have a perfectly good notion of length: the length of the interval $[a, b]$ is $b - a$. What could go wrong?

It turns out that when we try to build a theory of probability on the real line, we need to assign a "size" (a probability, or more generally a measure) not just to intervals, but to complicated sets formed by taking countably many unions, intersections, and complements of intervals. The naive approach—just add up the lengths of the pieces—runs into serious problems once we allow infinitely many operations.

> **Motivating example.**
>
> Consider the set $\mathbb{Q} \cap [0, 1]$, the rational numbers in $[0, 1]$. This set is *dense* (between any two points there is a rational), so it "looks" as if it should have positive length. Yet there are only *countably many* rationals, and each single point has length zero. Intuitively, infinitely many zeros should still sum to zero. Lebesgue measure will confirm this: $\lambda(\mathbb{Q} \cap [0, 1]) = 0$, even though the rationals are dense. The Riemann integral, by contrast, cannot even "integrate" the indicator function of $\mathbb{Q}$ over $[0, 1]$.

### 8.1.2 Building Lebesgue Measure Step by Step

Think of Lebesgue measure $\lambda$ as the "correct" generalisation of length from intervals to a much larger collection of sets. We build it in three intuitive steps.

#### 8.1.2.1 Step 1 — Start with intervals

For any interval (open, closed, or half-open), declare its measure to equal its length:

$$\lambda([a, b]) \;=\; \lambda((a, b)) \;=\; b - a.$$

Single points have measure zero: $\lambda(\{a\}) = a - a = 0$.

#### 8.1.2.2 Step 2 — Cover complicated sets with intervals

For a general set $E \subseteq \mathbb{R}$, we define the *outer measure* $\lambda^*(E)$ by trying to cover $E$ with a countable collection of open intervals $\{I_1, I_2, \ldots\}$ and asking: what is the smallest total length we can achieve?

$$\lambda^*(E) \;=\; \inf\left\{ \sum_{n=1}^{\infty} |I_n| \;\middle|\; E \subseteq \bigcup_{n=1}^{\infty} I_n,\; I_n \text{ open intervals} \right\}.$$

> **Geometric picture.**
>
> Imagine draping a "net" of open intervals over the set $E$. Outer measure is the least amount of net you must use, optimised over all possible covers. For an interval $[a, b]$ the optimum cover is just $(a - \varepsilon, b + \varepsilon)$, and shrinking $\varepsilon \to 0$ recovers $\lambda^*([a, b]) = b - a$. For a single point the cover can be made arbitrarily thin, giving $\lambda^*(\{a\}) = 0$.

#### 8.1.2.3 Step 3 — Identify the "well-behaved" sets (measurable sets)

Not every subset of $\mathbb{R}$ behaves nicely under this covering procedure. A set $E$ is called *Lebesgue measurable* if it does not "straddle" any other set in a pathological way; the precise condition (Carathéodory's criterion) is:

$$\lambda^*(A) \;=\; \lambda^*(A \cap E) \;+\; \lambda^*(A \cap E^c) \quad \text{for every set } A \subseteq \mathbb{R}.$$

For measurable sets, we simply write $\lambda(E) := \lambda^*(E)$ and call it the *Lebesgue measure* of $E$.

The collection $\mathcal{L}$ of all Lebesgue measurable sets is enormous. It contains: all open sets, all closed sets, all countable unions and intersections thereof (the Borel sets), and many more. In practice, every set you will ever encounter in economics or probability is Lebesgue measurable.

**Definition (Lebesgue Measure)** The *Lebesgue measure* on $\mathbb{R}$ is the function

$$\lambda : \mathcal{L} \to [0, \infty]$$

satisfying:

1. $\lambda([a,b]) = b - a$ for all $a \leq b$.
2. Countable additivity: if $E_1, E_2, \ldots$ are pairwise disjoint measurable sets, then $\lambda\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \lambda(E_n)$.
3. Translation invariance: $\lambda(E + t) = \lambda(E)$ for all $t \in \mathbb{R}$.

### 8.1.3 Key Properties of Lebesgue Measure

**Countable sets have measure zero.** If $E = \{x_1, x_2, x_3, \ldots\}$ is a countable set, then by countable additivity, $\lambda(E) = \sum_{n=1}^{\infty} \lambda(\{x_n\}) = \sum_{n=1}^{\infty} 0 = 0$. In particular, $\lambda(\mathbb{Q}) = 0$: the rational numbers, despite being dense in $\mathbb{R}$, occupy no space at all in the sense of Lebesgue measure.

**Measure is monotone.** If $A \subseteq B$ and both are measurable, then $\lambda(A) \leq \lambda(B)$.

**Countable subadditivity.** For any sequence of measurable sets (not necessarily disjoint), $\lambda\left(\bigcup_n E_n\right) \leq \sum_n \lambda(E_n)$.

**Continuity of measure.** If $E_1 \subseteq E_2 \subseteq \cdots$ is an increasing sequence with union $E$, then $\lambda(E_n) \to \lambda(E)$; if $E_1 \supseteq E_2 \supseteq \cdots$ is decreasing with $\lambda(E_1) < \infty$, then $\lambda(E_n) \to \lambda\left(\bigcap_n E_n\right)$.

### 8.1.4 The Lebesgue Integral

The central insight behind Lebesgue integration is to change what we partition. Rather than slicing the *domain* (the $x$-axis, as Riemann does), we slice the *range* (the $y$-axis) and ask how much of the domain maps into each slice. Figure~**??** illustrates this contrast.

#### 8.1.4.1 Step 1 — Simple functions

A *simple function* takes only finitely many values. If $\phi = \sum_{i=1}^{n} a_i \, 1_{E_i}$ where the $E_i$ are disjoint measurable sets and $a_i \geq 0$, its Lebesgue integral over a measurable set $A$ is defined as the natural weighted sum:

$$\int_A \phi \, d\lambda \;=\; \sum_{i=1}^{n} a_i \, \lambda(E_i \cap A).$$

> **Analogy.**
>
> Think of a simple function as a salary schedule that pays $a_i$ dollars per hour to employees in group $E_i$. The integral is the total wage bill: rate ×size-of-group, summed over all groups. Lebesgue measure plays the role of "group size."

#### 8.1.4.2 Step 2 — Non-negative measurable functions

For a non-negative measurable function $f \geq 0$, we approximate $f$ from below by an increasing sequence of simple functions $\phi_1 \leq \phi_2 \leq \cdots \uparrow f$ and define:

$$\int_A f \, d\lambda \;=\; \lim_{n\to\infty} \int_A \phi_n \, d\lambda.$$

#### 8.1.4.3 Step 3 — General measurable functions

Any measurable function can be split into its positive and negative parts: $f = f^+ - f^-$, where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$. Both $f^+$ and $f^-$ are non-negative, so Step 2 applies to each. We then define:

$$\int_A f \, d\lambda \;=\; \int_A f^+ \, d\lambda \;-\; \int_A f^- \, d\lambda,$$

provided at least one of the two terms is finite. If both are finite, $f$ is called *Lebesgue integrable* and we write $f \in L^1(\lambda)$.

### 8.1.5 What Makes Lebesgue Integration Better?

The Lebesgue integral agrees with the Riemann integral whenever the Riemann integral exists — so nothing is lost. But the Lebesgue integral handles many situations the Riemann integral cannot. Three properties are especially important for probability theory.

1. Integrable functions can be wildly discontinuous. The indicator function $1_{\mathbb{Q}}$ is not Riemann integrable over $[0, 1]$. Its Lebesgue integral is 0.
2. "Almost everywhere" is the right notion of equality. We say two functions $f$ and $g$ are equal *almost everywhere* (a.e.) if $\lambda(\{x : f(x) \neq g(x)\}) = 0$.
3. Limit and integral can be swapped under mild conditions (Monotone Convergence, Fatou's Lemma, Dominated Convergence).

### 8.1.6 Connection to Probability

Everything above extends directly to probability spaces. A *probability measure* $P$ on a measurable space $(\Omega, \mathcal{F})$ is simply a measure with $P(\Omega) = 1$. The Lebesgue integral of a random variable $X$ with respect to $P$ is its expected value:

$$\mathbb{E}[X] \;=\; \int_{\Omega} X(\omega)\, dP(\omega).$$

For a thorough treatment at the graduate level, see Royden and Fitzpatrick *Real Analysis* (4th ed.) and Billingsley *Probability and Measure* (3rd ed.).

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